Formulação ILP - By Intervals

Hugo Dias

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1 Common Constraints

Hard Constraints (must be satisfied):

- **One work block per day**: Each employee works at most one work block per day.
- **No work during vacations**: Employees cannot work on days they are on vacation.
- **Team authorization**: Each employee can only work in eligible teams.
- **Exactly 223 working days**: Each employee must work exactly 223 days per year.
- **Maximum 5 consecutive days**: No employee can work more than 5 consecutive days in any 6-day window.
- **Minimum 12 Hours rest period**: Employees must have at least 12 hours of rest between consecutive work days.
- **No work on closed days**: Employees cannot work on closed days.

Soft Constraints (violations penalized in objective):

- **Minimum coverage requirements**: For each day, hour, and team, the coverage must meet the minimum requirement.

2 Integer Linear Programming Model (Version 2)

2.1 Set of Indices

Set	Description
E	Set of employees
D	Set of days, $D = \{1, 2, \dots, \lvert D\rvert\}$, where $\lvert D\rvert = \begin{cases} 365, & \text{if common year} \\ 366, & \text{if leap year} \end{cases}$
B	Set of possible work blocks
H	Set of store operating hours
T	Set of teams
$C \subset D$	Set of closed days
$A_r \subset B$	Set of blocks whose assignment to a employee depends on the block assigned to the same employee in the previous day
$B_a \subset B$	Set of blocks $b \in B$ such that if one of them is assigned to an employee in one day, block a cannot be assigned to the same employee on the next day

2.2 Parameters

Parameter	Description
θ_{dht}	Minimum number of employees required on day d, hour h, team t
δ_{ed}	1 if employee e is on vacation on day d, 0 otherwise
α_{bh}	1 if block b covers hour h, 0 otherwise
T_e	Set of teams t that employee e can work

2.3 Decision Variables

Variable	Description
$x_{edbt} \in \{0, 1\}$	1 if employee e works on day d, block b, team $t \in T_e$
$s_{dht} \in \mathbb{Z}^+$	Number of employees working on day d, hour h, team t

2.4 Objective Function

Minimize the sum of the employee's shortage over all days d, all time periods h and all teams t

$$\min \sum_{d \in D} \sum_{h \in H} \sum_{t \in T} s_{dht} \tag{1}$$

2.5 Constraints

One work block per day and no work during vacations

$$\sum_{b \in B} \sum_{t \in T_e} x_{edbt} \leq 1 - \delta_{ed} \quad e \in E, d \in D \tag{2}$$

For each employee $e \in E$ on each day $d \in D$, either a single working block $b \in B$ can be assigned if e is not on vacation in d (i.e., if $\delta_{ed} = 0$) or no working block can be assigned if e is on vacation in d (i.e., if $\delta_{ed} = 1$).

Team authorization

This constraint is satisfied when we create the x_{edbt} variable

Exactly 223 working days

$$\sum_{d \in D} \sum_{b \in B} \sum_{t \in T_e} x_{edbt} = 223 \quad e \in E \tag{3}$$

Each employee $e \in E$ must be assigned with 223 working days.

Maximum 5 consecutive days

$$\sum_{d=d'}^{d'+5} \sum_{b \in B} \sum_{t \in T_e} x_{ed'bt} \leq 5 \quad e \in E, d' = 1, ..., |D| - 5 \tag{4}$$

For every sequence of 6 consecutive days, the number of assigned working days to each employee $e \in E$ can be at most 5.

Minimum 12 Hours rest period

$$\sum_{b \in B_a} \sum_{t \in T_e} x_{edbt} + \sum_{t \in T_e} x_{e,d+1,at} \leq 1 \quad e \in E, a \in A, d = 1, ..., |D| - 1 \tag{5}$$

For each employee $e \in E$ and each block $a \in A_r$ (i.e., the working day blocks whose assignment depends on the day before), if one of the blocks in B_a is assigned in one day (i.e., if $\sum_{b \in B_a} \sum_{t \in T_a} x_{edbt} = 1$), than, block a cannot be assigned in day $d + 1$.

No work on closed days

$$\sum_{e \in E} \sum_{d \in C} \sum_{b \in B} \sum_{t \in T_e} x_{fdbt} = 0 \tag{6}$$

No employees work on closed days

Minimum coverage requirements

$$\sum_{e \in E} \sum_{b \in B} x_{edbt} \cdot \alpha_{bh} + s_{dht} \geq \theta_{dht} \quad d \in D, h \in H, t \in T \tag{7}$$

These constraints (together with the objective function) guarantee that variable s_{dht} has the shortage of employees for each day $d \in D$, each time period $h \in H$ and each team $t \in T$.

3 Integer Linear Programming Model (Version 3)

3.1 Index Sets

Set	Description
E	Set of employees, $e \in E$
D	Set of days, $d = 1, 2, \ldots, \lvert D\rvert$, where $\lvert D\rvert = \begin{cases} 365, & \text{if common year} \\ 366, & \text{if leap year} \end{cases}$
B	Set of possible (working day) blocks, $b \in B$
H	Set of time periods of a day, $h \in H$
T	Set of teams, $t \in T$
$A \subset B$	Set of blocks whose assignment to a employee depends on the block assigned to the same employee in the previous day
$B_a \subset B$	Set of blocks $b \in B$ such that if one of them is assigned to an employee in one day, block $a \in A$ cannot be assigned to the same employee on the next day

3.2 Parameters

Parameter	Description
λ_d	1 if day d is a closed day, otherwise 0
θ_{dht}	Minimum number of employees required on day d, time period h, team t (these values are 0 for close days, i.e,, for days d such that $\lambda_d = 0$)
δ_{ed}	1 if employee e is on vacation on day d, 0 otherwise
α_{bh}	1 if block b covers time period h, 0 otherwise
T_e	Set of teams t that employee e can work

3.3 Decision Variables

Variable	Description
$x_{edbt} \in \{0, 1\}$	1 if employee e works on day d, block b, team t
$z_{edb} \in \{0, 1\}$	1 if employee e works on day d in block b
$s_{dht} \in \mathbb{Z}_0^+$	Shortage of employees relative to the required minimum on day d, hour h, team t

3.4 Objective Function

Minimize the sum of the employee's shortage over all days d that are not close days (i.e., $\lambda_d = 1$), all time periods h and all teams t

$$\min \sum_{d \in D} \sum_{h \in H} \sum_{t \in T} \lambda_d s_{dht} \tag{1}$$

3.5 Constraints

One work block per day and no work during vacations

$$\sum_{b \in B} z_{edb} \leq 1 - \delta_{ed} \qquad e \in E, d \in D \tag{2}$$

For each employee $e \in E$ on each day $d \in D$, either a single working block $b \in B$ can be assigned if e is not on vacation in d (i.e., if $\delta_{ed} = 0$) or no working block can be assigned if e is on vacation in d (i.e., if $\delta_{ed} = 1$).

Team authorization

$$x_{edbt} = 0 \qquad \forall e \in E, d \in D, b \in B, t \notin T_e \tag{3}$$

No employee $e \in E$ can be assigned with any working block $b \in B$ in any day $d \in D$ in a team $t \in T$ not belonging to T_e.

Exactly 223 working days

$$\sum_{d \in D} \sum_{b \in B} z_{edb} = 223 \quad e \in E \tag{4}$$

Each employee $e \in E$ must be assigned with 223 working days.

Maximum 5 consecutive days

$$\sum_{d=d'}^{d'+5} \sum_{b \in B} z_{edb} \leq 5 \quad e \in E,\ d' = 1, \ldots, |D| - 5 \tag{5}$$

For every sequence of 6 consecutive days, the number of assigned working days to each employee $e \in E$ can be at most 5.

Minimum 12-hour rest period

$$\sum_{b \in B_a} \sum_{t \in T_a} x_{edbt} + \sum_{t \in T_a} x_{e,d+1,at} \leq 1 \quad e \in E, a \in A, d = 1, ..., |D| - 1 \tag{6}$$

For each employee $e \in E$ and each block $a \in A_r$ (i.e., the working day blocks whose assignment depends on the day before), if one of the blocks in B_a is assigned in one day (i.e., if $\sum_{b \in B_a} \sum_{t \in T_a} x_{edbt} = 1$), than, block a cannot be assigned in day $d + 1$.

Minimum coverage requirements

$$s_{dht} \geq \theta_{dht} - \sum_{e \in E} \sum_{b \in B} \alpha_{bh}\, x_{edbt} \quad d \in \{D : \lambda_d = 0\}, h \in H, t \in T \tag{7}$$

These constraints (together with the objective function) guarantee that variable s_{dht} has the shortage of employees for each non-closed day $d \in \{D : \lambda_d = 0\}$, each time period $h \in H$ and each team $t \in T$.

Linking constraints between variables x and z

$$\sum_{T \in T_e} x_{edbt} = z_{edb} \quad e \in E, d \in D, b \in B \tag{8}$$

This constraint is the foundation of the variable z, which is a simplification of the team's sum of x. It could be removed if we used only the variable x in the other constraints , instead of z.

4 Integer Linear Programming Model (Version 4)

4.1 Set of Indices

Set	Description
E	Set of employees
D	Set of days, $D = \{1, 2, \ldots, \lvert D \rvert\}$, where $\lvert D \rvert = \begin{cases} 365, & \text{if common year} \\ 366, & \text{if leap year} \end{cases}$
B	Set of possible work blocks
H	Set of store operating hours
T	Set of teams
$C \subset D$	Set of closed days
$A_r \subset B$	Set of blocks whose assignment to a employee depends on the block assigned to the same employee in the previous day
$B_a \subset B$	Set of blocks $b \in B$ such that if one of them is assigned to an employee in one day, block a cannot be assigned to the same employee on the next day

4.2 Parameters

Parameter	Description
θ_{dht}	Minimum number of employees required on day d, hour h, team t
β_{dht}	Ideal number of employees required on day d, hour h, team t
δ_{ed}	1 if employee e is on vacation on day d, 0 otherwise
α_{bh}	1 if block b covers hour h, 0 otherwise
T_e	Set of teams t that employee e can work

4.3 Decision Variables

Variable	Description
$x_{edbt} \in \{0, 1\}$	1 if employee e works on day d, block b, team $t \in T_e$
$s_{dht} \in \mathbb{Z}^+$	Number of employees working on day d, hour h, team t
$z_{dht} \in \mathbb{Z}^+$	Non-negative variable with the number of workers below the ideal value β_{dht} on day d, hour h and team t

4.4 Objective Function

s_{dht} minimize the sum of the employee's shortage over all days d, all time periods h and all teams t. If this shortage is satisfied, then the model try to minimize the ideal values z_{dht} also over all days d, all time periods h and all teams t. This condition is represented by the weight 10000, which prioritizes the minimums over the ideals

$$\min \left(\sum_{d \in D} \sum_{h \in H} \sum_{t \in T} z_{dht} + 10000 \cdot \sum_{d \in D} \sum_{h \in H} \sum_{t \in T} s_{dht} \right) \tag{1}$$

4.5 Constraints

One work block per day and no work during vacations

$$\sum_{b \in B} \sum_{t \in T_e} x_{edbt} \leq 1 - \delta_{ed} \quad e \in E, d \in D \tag{2}$$

For each employee $e \in E$ on each day $d \in D$, either a single working block $b \in B$ can be assigned if e is not on vacation in d (i.e., if $\delta_{ed} = 0$) or no working block can be assigned if e is on vacation in d (i.e., if $\delta_{ed} = 1$).

Team authorization

This constraint is satisfied when we create the x_{edbt} variable

Exactly 223 working days

$$\sum_{d \in D} \sum_{b \in B} \sum_{t \in T_e} x_{edbt} = 223 \quad e \in E \tag{3}$$

Each employee $e \in E$ must be assigned with 223 working days.

Maximum 5 consecutive days

$$\sum_{d=d'}^{d'+5} \sum_{b \in B} \sum_{t \in T_e} x_{ed'bt} \leq 5 \quad e \in E, d' = 1, ..., |D| - 5 \tag{4}$$

For every sequence of 6 consecutive days, the number of assigned working days to each employee $e \in E$ can be at most 5.

Minimum 12 Hours rest period

$$\sum_{b \in B_a} \sum_{t \in T_e} x_{edbt} + \sum_{t \in T_e} x_{e,d+1,at} \leq 1 \quad e \in E, a \in A, d = 1, ..., |D| - 1 \tag{5}$$

For each employee $e \in E$ and each block $a \in A_r$ (i.e., the working day blocks whose assignment depends on the day before), if one of the blocks in B_a is assigned in one day (i.e., if $\sum_{b \in B_a} \sum_{t \in T_a} x_{edbt} = 1$), than, block a cannot be assigned in day $d + 1$.

No work on closed days

$$\sum_{e \in E} \sum_{d \in C} \sum_{b \in B} \sum_{t \in T_e} x_{fdbt} = 0 \tag{6}$$

No employees work on closed days

Minimum coverage requirements

$$\sum_{e \in E} \sum_{b \in B} x_{edbt} \cdot \alpha_{bh} + s_{dht} \geq \theta_{dht} \quad d \in D, h \in H, t \in T \tag{7}$$

These constraints (together with the objective function) guarantee that the variable s_{dht} represents the shortage relative to the minimum required number of employees for each day $d \in D$, each time period $h \in H$ and each team $t \in T$.

Ideal coverage requirements

$$\sum_{e\in E}\sum_{b\in B} x_{edbt} \cdot \alpha_{bh} + z_{dht} \geq \beta_{dht} \quad d \in D, h \in H, t \in T \tag{8}$$

These constraints (together with the objective function) guarantee that the variable s_{dht} represents the shortage relative to the ideal required number of employees for each day $d \in D$, each time period $h \in H$ and each team $t \in T$

5 Constraint Optimization Model (Version 1)

5.1 Sets and Indices

Set	Description
E	Set of employees
D	Set of days, $D = \{1, 2, \ldots, \lvert D\rvert\}$, where $\lvert D\rvert = \begin{cases} 365, & \text{if common year} \\ 366, & \text{if leap year} \end{cases}$
B	Set of possible work blocks
H	Set of store operating hours
T	Set of teams
$V \subset D$	Set of vacation days
$A_r \subset B$	Set of blocks whose assignment to a employee depends on the block assigned to the same employee in the previous day
$B_a \subset B$	Set of blocks $b \in B$ such that if one of them is assigned to an employee in one day, block a cannot be assigned to the same employee on the next day

5.2 Parameters

Parameter	Description
θ_{dht}	Minimum number of employees required on day d, hour h, team t
α_{bh}	1 if block b covers hour h, 0 otherwise
λ_d	1 if day d is a closed day, otherwise 0
T_e	Set of teams t that employee e can work

5.3 Decision Variables

Variable	Description
$x_{edbt} \in \{0, 1\}$	1 if employee e works on day d, block b, team $t \in T_e$
$w_{ed} \in \{0, 1\}$	1 if employee e works on day d
$s_{dht} \in \mathbb{Z}^+$	Number of employees working on day d, hour h, team t

5.4 Objective Function

Minimize the sum of the employee's shortage over all days d that are not close days (i.e., $\lambda_d = 1$), all time periods h and all teams t

$$\min \sum_{d\in D} \sum_{h\in H} \sum_{t\in T} \lambda_d s_{dht} \tag{1}$$

5.5 Constraints

One work block per day

$$\sum_{b\in B} \sum_{t\in T_e} x_{edbt} \leq 1 \quad e \in E, d \in D \tag{2}$$

For each employee $e \in E$ on each day $d \in D$, only a single working block $b \in B$ can be assigned

No work during vacations

$$w_{ed} = 0 \quad e \in E, d \in V \tag{3}$$

For each employee $e \in E$ on each vacation day $d \in V$, no working block $b \in B$ can be assigned

Exactly 223 working days

$$\sum_{d \in D} w_{ed} = 223 \quad e \in E \tag{4}$$

Each employee $e \in E$ must be assigned with 223 working days.

Maximum 5 consecutive days

$$\sum_{d=d'}^{d'+5} w_{ed} \leq 5 \quad e \in E,\ d' = 1, \ldots, |D| - 5 \tag{5}$$

For every sequence of 6 consecutive days, the number of assigned working days to each employee $e \in E$ can be at most 5.

Minimum 12 Hours rest period

$$\sum_{b \in B_a} \sum_{t \in T_e} x_{edbt} + \sum_{t \in T_e} x_{e,d+1,at} \leq 1 \quad e \in E, a \in A, d = 1, ..., |D| - 1 \tag{6}$$

For each employee $e \in E$ and each block $a \in A_r$ (i.e., the working day blocks whose assignment depends on the day before), if one of the blocks in B_a is assigned in one day (i.e., if $\sum_{b \in B_a} \sum_{t \in T_a} x_{edbt} = 1$), than, block a cannot be assigned in day $d + 1$.

Minimum coverage requirements

$$s_{dht} \geq \theta_{dht} - \sum_{e \in E} \sum_{b \in B} \alpha_{bh}\, x_{edbt} \quad d \in d, h \in H, t \in T \tag{7}$$

These constraints (together with the objective function) guarantee that the variable s_{dht} represents the shortage relative to the minimum required number of employees for each day $d \in D$, each time period $h \in H$ and each employee team $t \in T$.

Definition of working day

$$w_{ed} = \sum_{b \in B} \sum_{e \in T_e} x_{edbt} \quad e \in E, d \in D \tag{8}$$

This constraint is the foundation of the variable w, which is a simplification of the Team's sum and Block's sum of x. It could be removed if we used only the variable x in the other constraints , instead of w.

6 Constraint Optimization Model (Version 2)

6.1 Set of Indices

Set	Description
E	Set of employees
D	Set of days, $D = \{1, 2, \ldots, \lvert D\rvert\}$, where $\lvert D\rvert = \begin{cases} 365, & \text{if common year} \\ 366, & \text{if leap year} \end{cases}$
B	Set of possible work blocks
H	Set of store operating hours
T	Set of teams
$C \subset D$	Set of closed days
$A_r \subset B$	Set of blocks whose assignment to a employee depends on the block assigned to the same employee in the previous day
$B_a \subset B$	Set of blocks $b \in B$ such that if one of them is assigned to an employee in one day, block a cannot be assigned to the same employee on the next day

6.2 Parameters

Parameter	Description
θ_{dht}	Minimum number of employees required on day d, hour h, team t
δ_{ed}	1 if employee e is on vacation on day d, 0 otherwise
α_{bh}	1 if block b covers hour h, 0 otherwise
T_e	Set of teams t that employee e can work

6.3 Decision Variables

Variable	Description
$x_{edbt} \in \{0, 1\}$	1 if employee e works on day d, block b, team $t \in T_e$
$s_{dht} \in \mathbb{Z}^+$	Number of employees working on day d, hour h, team t

6.4 Objective Function

Minimize the sum of the employee's shortage over all days d, all time periods h and all teams t

$$\min \sum_{d \in D} \sum_{h \in H} \sum_{t \in T} s_{dht} \tag{1}$$

6.5 Constraints

One work block per day and no work during vacations

$$\sum_{b \in B} \sum_{t \in T_e} x_{edbt} \leq 1 - \delta_{ed} \quad e \in E, d \in D \tag{2}$$

For each employee $e \in E$ on each day $d \in D$, either a single working block $b \in B$ can be assigned if e is not on vacation in d (i.e., if $\delta_{ed} = 0$) or no working block can be assigned if e is on vacation in d (i.e., if $\delta_{ed} = 1$).

Team authorization

This constraint is satisfied when we create the x_{edbt} variable

Exactly 223 working days

$$\sum_{d \in D} \sum_{b \in B} \sum_{t \in T_e} x_{edbt} = 223 \quad e \in E \tag{3}$$

Each employee $e \in E$ must be assigned with 223 working days.

Maximum 5 consecutive days

$$\sum_{d=d'}^{d'+5} \sum_{b \in B} \sum_{t \in T_e} x_{ed'bt} \leq 5 \quad e \in E, d' = 1, ..., |D| - 5 \tag{4}$$

For every sequence of 6 consecutive days, the number of assigned working days to each employee $e \in E$ can be at most 5.

Minimum 12 Hours rest period

$$\sum_{b \in B_a} \sum_{t \in T_e} x_{edbt} + \sum_{t \in T_e} x_{e,d+1,at} \leq 1 \quad e \in E, a \in A, d = 1, ..., |D| - 1 \tag{5}$$

For each employee $e \in E$ and each block $a \in A_r$ (i.e., the working day blocks whose assignment depends on the day before), if one of the blocks in B_a is assigned in one day (i.e., if $\sum_{b \in B_a} \sum_{t \in T_a} x_{edbt} = 1$), than, block a cannot be assigned in day $d + 1$.

No work on closed days

$$\sum_{e \in E} \sum_{d \in C} \sum_{b \in B} \sum_{t \in T_e} x_{fdbt} = 0 \tag{6}$$

No employees work on closed days

Minimum coverage requirements

$$\sum_{e \in E} \sum_{b \in B} x_{edbt} \cdot \alpha_{bh} + s_{dht} \geq \theta_{dht} \quad d \in D, h \in H, t \in T \tag{7}$$

These constraints (together with the objective function) guarantee that variable s_{dht} has the shortage of employees for each day $d \in D$, each time period $h \in H$ and each team $t \in T$.